Browser Forensic Artifact Analysis Report

Author: S.M.A.D.R.Pabasarani

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Tools Used: DB Browser for SQLite, Hindsight, Google Chrome

Platform: Windows 10

1. Objective

The objective of this project was to perform digital forensic analysis on Google Chrome browser artifacts. The goal was to extract and examine browsing history, downloads, search queries, and other artifacts to reconstruct user activity. This project demonstrates foundational skills in digital forensics and incident response (DFIR).

2. Tools Used

- DB Browser for SQLite – Used to manually inspect the Chrome History database file and examine the urls table.
- Hindsight – An open-source forensic tool that automatically parses Chrome browser data and generates a timeline of user activity.
- Google Chrome – Source of browser artifacts.
- Windows OS – Host system for analysis.

3. Methodology

3.1 Evidence Acquisition

- Google Chrome was completely closed to unlock database files.
- The Chrome profile folder was located at: C:\Users\[Username]\AppData\Local\Google\Chrome\User Data\Default
- The History file was copied to the desktop to preserve the original evidence.

3.2 Manual Analysis with DB Browser for SQLite

- The copied History file was opened in DB Browser for SQLite.
- The Browse Data tab was used to examine the urls table.
- The following columns were reviewed:
    - url – website addresses
    - title – page titles
    - visit_count – number of visits
    - last_visit_time – raw Chrome timestamp (microseconds since 1601-01-01)

3.3 Automated Analysis with Hindsight

- Hindsight GUI was launched by running hindsight-gui.exe.
- A browser window opened automatically at http://localhost:8080.
- In the web interface, the Chrome profile path was entered: C:\Users\[YourName]\AppData\Local\Google\Chrome\User Data\Default
- The Run button was clicked, and Hindsight processed the profile.
- After completion, the results page displayed a summary of extracted artifacts, including URLs, downloads, search queries, bookmarks, and more.

3.4 Timeline Reconstruction

- Both manual (DB Browser) and automated (Hindsight) results were reviewed to identify browsing patterns.
- Frequently visited websites were noted from the urls table in DB Browser.
- The Hindsight summary provided counts of various artifact types, helping to understand the scope of browsing activity.
- A general timeline of activity was inferred from the types of websites visited (e.g., educational, entertainment, professional).

4. Findings

4.1 Browser Artifacts Extracted (from Hindsight)

Artifact Type	Count
URL records	18,182
Download records	650
Google search queries	6,790
Bookmarks	182
Autofill entries	578

4.2 Browsing Patterns (from DB Browser and Hindsight)

- Frequently visited websites included:
    - Video streaming (e.g., YouTube, Netflix)
    - Educational platforms (e.g., university portals)
    - Professional networks (e.g., LinkedIn)
- Search queries indicated a mix of academic research, entertainment, and general browsing.
- Download records showed various file types (PDFs, images, etc.).

4.3 Activity Summary

Based on the extracted artifacts, the user's browsing activity included:

Activity Type	Observations
Website visits	Mix of educational, entertainment, and professional sites
Searches	Thousands of Google searches across various topics
Downloads	Multiple files downloaded, including documents and media
Bookmarks	Saved websites for easy access
Autofill	Form data stored for repeated use

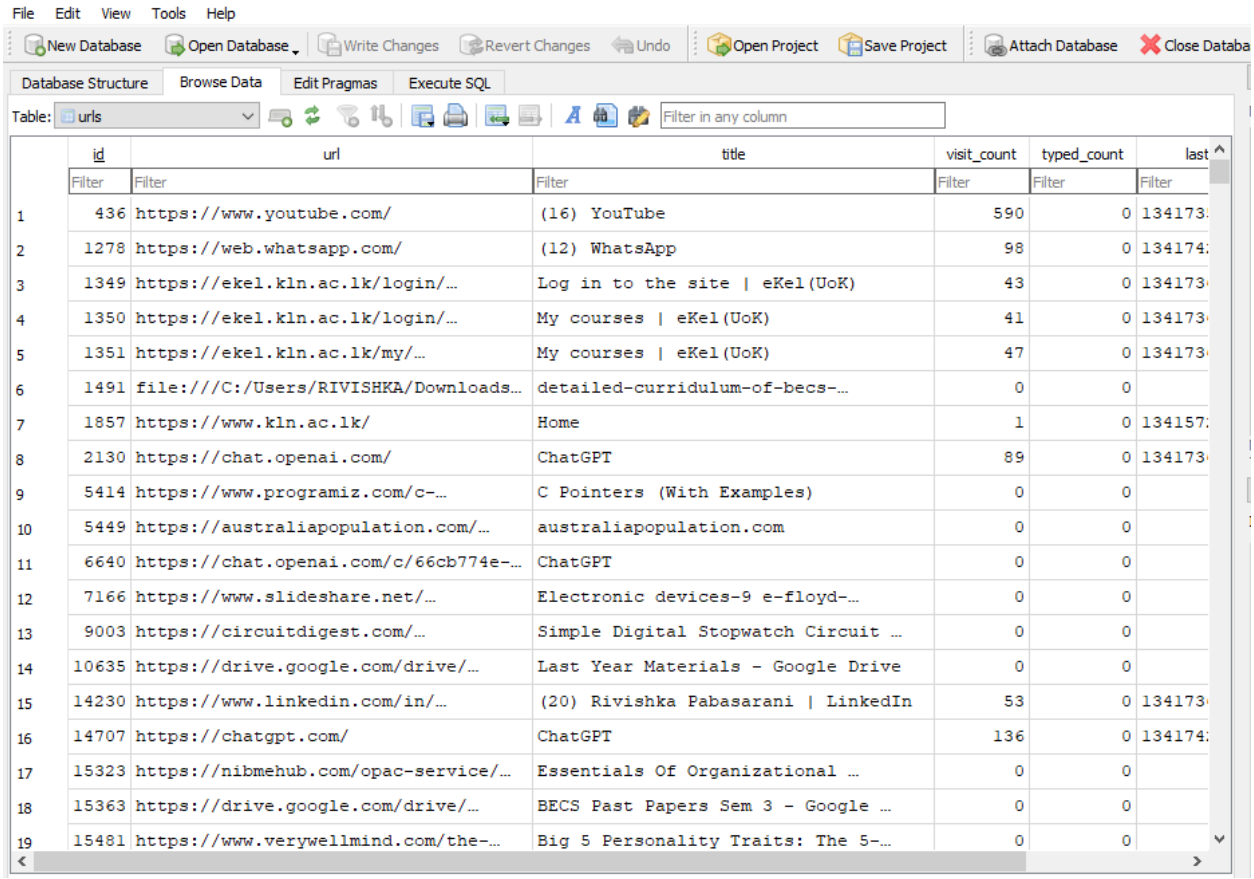
5. Conclusion

This project successfully demonstrated browser forensic analysis using both manual (DB Browser for SQLite) and automated (Hindsight GUI) techniques. Thousands of URL records, hundreds of downloads, and thousands of search queries were recovered, providing insight into user browsing behavior. The process highlighted the value of browser artifacts in digital

investigations and reinforced core DFIR workflows such as evidence acquisition, artifact analysis, and documentation.

Appendices

Appendix A: DB Browser – urls table view



File Edit View Tools Help

New Database | Open Database | Write Changes | Revert Changes | Undo | Open Project | Save Project | Attach Database | Close Databa

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Table: urls | Filter in any column

	id	url	title	visit_count	typed_count	last
1	436	https://www.youtube.com/	(16) YouTube	590	0	134173
2	1278	https://web.whatsapp.com/	(12) WhatsApp	98	0	134174
3	1349	https://ekel.kln.ac.lk/login/…	Log in to the site | eKel(UoK)	43	0	134173
4	1350	https://ekel.kln.ac.lk/login/…	My courses | eKel(UoK)	41	0	134173
5	1351	https://ekel.kln.ac.lk/my/…	My courses | eKel(UoK)	47	0	134173
6	1491	file:///C:/Users/RIVISHKA/Downloads…	detailed-curridulum-of-becs-…	0	0	
7	1857	https://www.kln.ac.lk/	Home	1	0	134157
8	2130	https://chat.openai.com/	ChatGPT	89	0	134173
9	5414	https://www.programiz.com/c-…	C Pointers (With Examples)	0	0	
10	5449	https://australiapopulation.com/…	australiapopulation.com	0	0	
11	6640	https://chat.openai.com/c/66cb774e-…	ChatGPT	0	0	
12	7166	https://www.slideshare.net/…	Electronic devices-9 e-floyd-…	0	0	
13	9003	https://circuitdigest.com/…	Simple Digital Stopwatch Circuit …	0	0	
14	10635	https://drive.google.com/drive/…	Last Year Materials - Google Drive	0	0	
15	14230	https://www.linkedin.com/in/…	(20) Rivishka Pabasarani | LinkedIn	53	0	134173
16	14707	https://chatgpt.com/	ChatGPT	136	0	134174
17	15323	https://nibmehub.com/opac-service/…	Essentials Of Organizational …	0	0	
18	15363	https://drive.google.com/drive/…	BECS Past Papers Sem 3 - Google …	0	0	
19	15481	https://www.verywellmind.com/the-…	Big 5 Personality Traits: The 5-…	0	0	

Appendix B: Hindsight – main results summary

Summary

Input Path: C:\Users\[redacted]\AppData\Local\Google\Chrome\User Data\Default

Input Type: Chrome

Profile Paths:

- C:\Users\[redacted]\AppData\Local\Google\Chrome\User Data\Default

Plugin Results

Chrome Extension Names [v20240428]:	- 35 extension URLs parsed -
Generic Timestamp Decoder [v20240428]:	- 0 timestamps parsed -
Google Analytics Cookie Parser [v20170130]:	- 0 cookies parsed -
Google Searches [v20160912]:	- 6790 searches parsed -
Load Balancer Cookie Decoder [v20200213]:	- 0 cookies parsed -
Quantcast Cookie Parser [v20160907]:	- 0 cookies parsed -
Query String Parser [v20170225]:	- 6336 query strings parsed -
Time Discrepancy Finder [v20170129]:	- 0 differences parsed -

Appendix C: Hindsight – plugins and extensions

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Profile
Path: C:\Users\[redacted]\AppData\Local\Google\Chrome\User Data\Default
Detected Browser: Chrome v142-144

User Activity                                Count
                          URL records    [   18182 ]
                     Download records    [     650 ]
                     Autofill records    [     578 ]
                  Login Data records     [      82 ]
          Login Data (Account) records   [       0 ]
                     Bookmark records    [     182 ]

Website Storage                              Count
                       Cookie records    [  Failed ]
                        Cache records    [  Failed ]
                    GPU Cache records    [  Failed ]
                Local Storage records    [   39825 ]
              Session Storage records    [     388 ]
                    IndexedDB records    [  135735 ]
                    File System items    [      24 ]

Browser Extensions                           Count
                  Installed Extensions    [       3 ]
             Extension Cookie records     [       0 ]
              Extension Rules records     [       0 ]
            Extension Scripts records     [   16840 ]
              Extension State records     [   35962 ]
     Local Extension Settings records     [    5277 ]
            Sync App Settings records     [       0 ]
        Sync Extension Settings records   [       0 ]

Configuration & Supporting Data              Count
                     Preference items     [    2125 ]
          Site Characteristics records    [   12070 ]
                    Sync Data records     [    3804 ]
                         HSTS records     [    2079 ]
                  DIPS Popup records      [      18 ]
                         DIPS records     [     256 ]

Running Plugins
```

```
Running Plugins

                 Chrome Extension Names (v20240428):   - 35 extension URLs parsed -
             Generic Timestamp Decoder (v20240428):     - 0 timestamps parsed -
       Google Analytics Cookie Parser (v20170130):       - 0 cookies parsed -
                     Google Searches (v20160912):     - 6790 searches parsed -
           Load Balancer Cookie Decoder (v20200213):       - 0 cookies parsed -
                Quantcast Cookie Parser (v20160907):       - 0 cookies parsed -
                    Query String Parser (v20170225):  - 6336 query strings parsed -
                Time Discrepancy Finder (v20170129):     - 0 differences parsed -
127.0.0.1 - - [08/Mar/2026 11:37:59] "POST /run HTTP/1.1" 303 0
127.0.0.1 - - [08/Mar/2026 11:38:00] "GET /results HTTP/1.1" 200 13518
127.0.0.1 - - [08/Mar/2026 11:38:00] "GET /static/styles.css HTTP/1.1" 304 0
127.0.0.1 - - [08/Mar/2026 11:38:00] "GET /static/github.png HTTP/1.1" 304 0
127.0.0.1 - - [08/Mar/2026 11:38:00] "GET /static/mastodon.svg HTTP/1.1" 304 0
127.0.0.1 - - [08/Mar/2026 11:38:00] "GET /static/h.png HTTP/1.1" 304 0
127.0.0.1 - - [08/Mar/2026 11:38:01] "GET /static/icons/manifest.json HTTP/1.1" 304 0
```